

Healthcare Appointment Management Platform

1. EPIC

Build and automate CI workflow for Healthcare Appointment Management Platform

2. FEATURES

2.1 Appointment Service API Development

2.1.1 User Story

Ensure service reliability using health check API

Tasks

- Create Flask application
- Implement "/" home endpoint
- Implement "/health" endpoint
- Test health endpoint locally

2.1.2 User Story

Check appointment status to know booking

Tasks

- Implement "/appointment-status" endpoint
 - Return appointment confirmation response
 - Validate API response
 - Test endpoint locally
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2.2 Doctor Availability Tracking

2.2.1 User Story

Doctor availability before booking an appointment

Tasks

- Implement "/doctor-availability" endpoint
 - Return doctor availability response
 - Validate API response
 - Test endpoint locally
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2.3 Automated Testing

2.3.1 User Story

Automated tests to validate API functionality

Tasks

- Install pytest
 - Write test cases for "/" endpoint
 - Write test cases for "/health" endpoint
 - Write test cases for "/appointment-status" endpoint
 - Write test cases for "/doctor-availability" endpoint
 - Run tests locally
 - Fix failing tests
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2.4 Repository Management (GitHub)

2.4.1 User Story

Version control to manage code changes efficiently

Tasks

- Create GitHub repository
 - Add project files (app.py, test_app.py, requirements.txt)
 - Commit code with proper messages
 - Push code to main branch
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2.5 CI Pipeline Setup (GitHub Actions)

2.5.1 User Story

CI pipeline automation for continuous integration

Tasks

- Create .github/workflows directory
- Add python-app.yml workflow file
- Configure Python setup in pipeline
- Install dependencies in pipeline
- Run pytest in pipeline

2.6 Build Verification and Validation

2.6.1 User Story

Build pass successfully before deployment

Tasks

- Trigger pipeline on code push
- Monitor GitHub Actions workflow
- Verify all tests pass
- Fix pipeline errors if any
- Confirm successful build execution

2.7 Application Monitoring

2.7.1 User Story

Monitor application health status

Tasks

- Use "/health" endpoint for monitoring
- Validate service availability
- Ensure correct response is returned